

Control Systems A

Week 1, Part B - Control System Components: a conceptual survey

UoN - FEE 3411

Sept - Dec 2026

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What this hour is

- A **survey** of the devices that fill the boxes in a block diagram. What each one does, how it does it, and roughly what its transfer function looks like.
- Transfer functions are **stated**, not derived. The derivations are in the full Week 1 notes, and the physics behind them is Week 2.

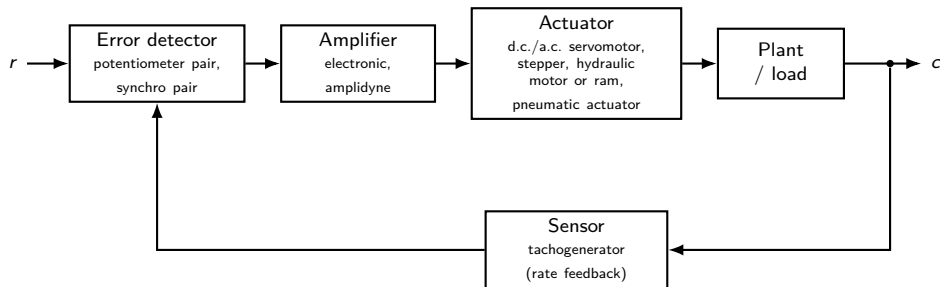
Why that is enough

The syllabus outcome here is ELO 1 — *“describe the working principles of a given control system”*. Describing is the requirement.

Companion document: `week-01-notes-partB-brief.pdf`. Full derivations: `week-01-notes.pdf`. Hardware source: Nagrath & Gopal 2nd ed., Ch. 4.

Where the components sit

The four families



Key idea

There are only four jobs in a control loop: **measure the error**, **amplify it**, **act on the plant**, **measure the output**. Every device today is a way of doing one of those four.

Meeting an unfamiliar component, the first question is: *which of the four is it?*

Error detectors

The potentiometer pair

Two potentiometers on a common d.c. supply — one turned by the reference shaft, one by the load shaft. The voltage between the wipers is

$$v_e = K_p (r - c), \quad [K_p] = \text{V rad}^{-1}.$$

The subtraction is done **by the wiring**, not by a circuit — this is the physical summing junction.

For

Cheap. A d.c. device. A **pure gain** with no dynamics at all.

Against

The wiper is a sliding contact: friction, wear, electrical noise, and a resolution set by the winding pitch.

The synchro pair — the idea

Where the shaft turns continuously, or lives somewhere hostile, a sliding contact is unacceptable. The **synchro** replaces it with transformer action — no contact anywhere in the signal path.

Key idea

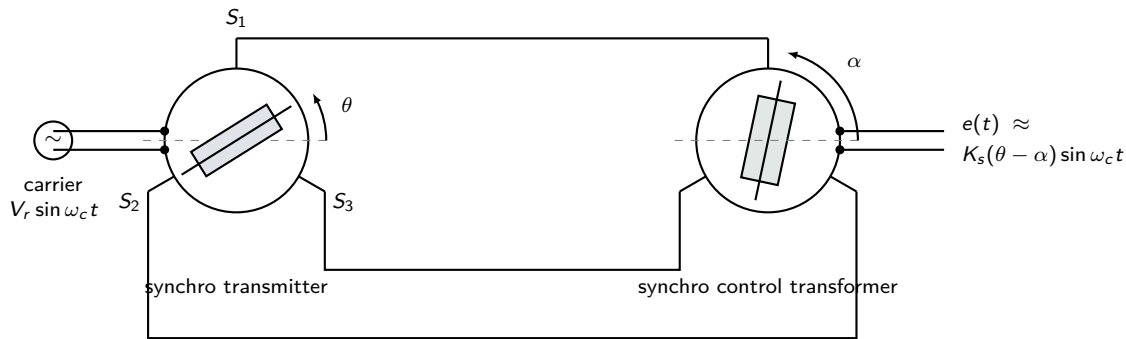
A synchro transmitter is a **transformer whose coupling depends on shaft angle**. Rotor coil = primary; three stator coils = secondaries; the voltage in each depends on its angle to the rotor.

In a shaft angle. **Out** three a.c. voltages that encode it.

Wire those three leads to a second machine — a **control transformer** — and its rotor voltage depends on the angle *between the two rotors*:

$$e(t) \approx K_s (\theta - \alpha) \sin \omega_c t.$$

The synchro pair as an error detector



K_s is the **sensitivity of the error detector**, $V \text{ rad}^{-1}$. No sliding contact anywhere in the signal path. Cf. Nagrath Fig. 4.12, p. 94.

Suppressed-carrier modulation

The output is a carrier $\sin \omega_c t$ whose **amplitude** carries the error and whose **phase flips** when the error changes sign.

A loop whose internal signals look like this is a **carrier control system**.

Common pitfall

The information is in the **envelope**, not the instantaneous value.

Read the error straight off an oscilloscope trace of $e(t)$ and you will conclude it is oscillating at 400 Hz — while the shaft stands perfectly still.

Actuators

The d.c. servomotor

A d.c. motor built for low inertia and rapid reversal.

- **Armature control** — field constant, armature voltage varied. Used almost universally.
- **Field control** — cheaper amplifier, but slower: the field winding is highly inductive.

$$\frac{\theta_m(s)}{V_a(s)} = \frac{K_m}{s(\tau_m s + 1)}$$

Read that shape now

A constant voltage gives a constant speed — hence the integrator, since position accumulates speed. The motor takes time to reach that speed — hence the first-order lag, because inertia and friction resist the change.

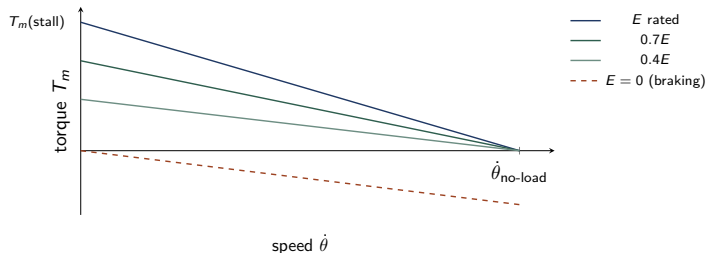
The a.c. (two-phase) servomotor

Preferred at low power, and wherever a synchro has already made the signals a.c. Rugged, light, no brushes. Structurally a two-phase induction motor. Two design changes make it a servomotor:

- 1 **High rotor resistance** — so the reactance-to-resistance ratio is *small*. In an ordinary induction motor it is deliberately large, which leaves part of the torque–speed curve *positively* sloped.
- 2 **Unbalanced excitation** — a fixed *reference phase* and a variable *control phase* at $\pm 90^\circ$ to it. Reverse the control voltage's sign and the motor reverses. Rotor is a small-diameter squirrel cage or drag cup, for low inertia.

Same shape as before: $\theta/E = K_m/[s(\tau_m s + 1)]$.

Why high rotor resistance? A stability argument



Key idea

The negative slope behaves like **extra viscous friction**: turn faster, get less torque, which damps the motor. A *positive* slope would behave like **negative friction** — and the loop could oscillate on its own. That is why an ordinary induction motor cannot be a servomotor: **stability reaches down into the choice of hardware.**

The stepper motor

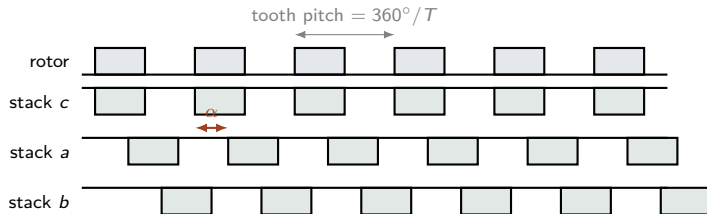
Converts a train of pulses into discrete angular steps — **one step per pulse**. The actuator of incremental motion control: printers, tape drives, machine tools, plotters, 3-D printers.

In a *variable-reluctance* stepper the toothed stator stacks are pulse excited and the toothed rotors are not. Energise a stack and the rotor is pulled to the nearest **minimum-reluctance** position — teeth aligned.

$$\alpha = \frac{360^\circ}{n T} \quad \begin{array}{l} n=\text{stacks (=phases)} \\ T=\text{rotor teeth} \end{array} \quad n = 3, T = 12 \Rightarrow \alpha = 10^\circ$$

abcabc ... steps one way, *bacbac* ... the other — **directional control needs three or more phases**.

Stepper motor — how the step happens



Rotor teeth are aligned across all stacks; stator stacks are offset successively by α . Energising c , then a , then b pulls the rotor one step at a time. Cf. Nagrath Figs. 4.18, 4.20, pp. 100–101.

Stepper motors: the trade

A digital actuator

Shaft position is determined **entirely by the pulse count**. So an *open-loop* step servo can reach the accuracy of a closed-loop analogue system with no position sensor at all — the rare case where open loop is the right engineering answer.

The price: skipped steps

If pulses arrive faster than the rotor can settle, a step is lost — and since nothing is measured, lost **silently and permanently**. High-performance systems close the loop after all.

Stepper dynamics are genuinely nonlinear and are solved numerically — one reason steppers do not reappear in the analysis weeks.

Sensors and rate feedback

The tachogenerator

$$v_t = K_t \dot{\theta}, \quad [K_t] = \text{V rad}^{-1} \text{s}^{-1}.$$

d.c. type

A small permanent-magnet generator.
Polarity follows the direction of rotation.
Commutation ripple is its main limitation.

a.c. drag-cup

A thin aluminium cup for a rotor, in the gap of a fixed magnetic structure. Very low inertia.

Used two ways: as the **sensor of a speed loop**; or as **rate feedback** inside a *position* loop, subtracted in an inner loop purely to add damping.

Rate feedback — your first compensator

Key idea

Adding rate feedback changes **only the damping** of the closed-loop system. It leaves the steady-state gain alone.

So you can calm an oscillatory servo *without making it less accurate*.

That is a **compensator**: a component added not because the plant needs it to function, but because the *loop* needs it to behave.

We use it as a design tool in Week 9. FEE3412 studies it properly. Full derivation — main notes §7; Nagrath §4.3, p. 96.

Hydraulic and pneumatic power

Why hydraulics?

For

- A hydraulic motor is far smaller than an electric one of the same power.
- Fast acting, rugged.
- Very stiff under load.

Against

- Leaks, sealing, contamination.
- Noise.
- Sluggish when cold.
- Inflexible lines.

Power steering and brakes, ship steering gear, machine tools, aircraft flight controls, excavators.

Rotary output = *hydraulic motors* — what the syllabus calls “rotary actuators”.

Translational = *linear actuators*, i.e. cylinders and rams.

Pump–motor transmission, and valve control

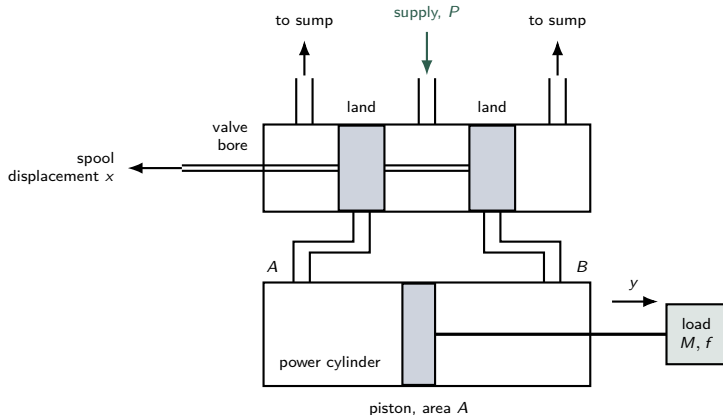
Pump–motor. A variable-stroke pump driving a fixed-stroke motor. Tilt the wobble plate through the *stroke angle* and oil is delivered in proportion; reverse the tilt and the motor reverses. One mechanical variable controls everything. Result: $\theta/X = K/[s(\tau s + 1)]$.

Valve control. Instead hold the supply pressure constant and throttle with a **spool valve**. Its moving part is far lighter than a pump's stroke mechanism, so time constants shrink and the system gets much faster.

The cost

Valve flow is nonlinear — through a sharp-edged orifice it goes as $\sqrt{\Delta p}$. The model we use is a **linearisation**, valid only for small spool movements about neutral.

Spool valve and power cylinder



Move the spool right: *B* opens to supply, *A* drains to sump, piston driven left. Move it the other way and the connections swap. The cylinder's transfer function is $Y/X = K/[s(\tau s + 1)]$. Cf. Nagrath Fig. 4.27, p. 111.

Where the $1/s$ comes from

The important idea in the whole hour

It is **not** an approximation, and it is not inertia.

Flow into a cylinder sets the piston's *velocity*. Position is therefore the **accumulation** of that velocity.

Any actuator whose input commands a rate — valve opening $\rightarrow$ velocity, armature voltage $\rightarrow$ speed, pump stroke $\rightarrow$ motor speed — carries a free integrator when the output you care about is position.

Pneumatics: air instead of oil

Non-inflammable, intrinsically safe, viscosity negligible and stable with temperature. But air is **compressible** — substantial compressibility flow, and **longer time delays**. Hence: *process* control, where the plant is slow anyway.

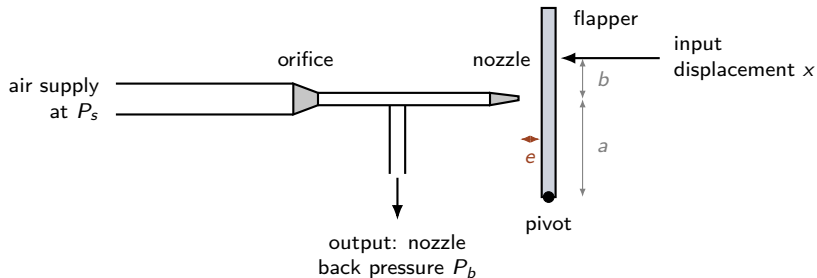
Bellows A corrugated chamber behaving as a spring. Pressure in, displacement out. A **pure gain**.

Flapper–nozzle valve Air through a fixed orifice and out of a nozzle; a pivoted flapper sits close by. Tiny movement, large pressure change — a high-gain displacement-to-pressure transducer.

Relay A pneumatic power amplifier in cascade, because the linear flapper range gives only a small pressure swing. Also inverts the sign.

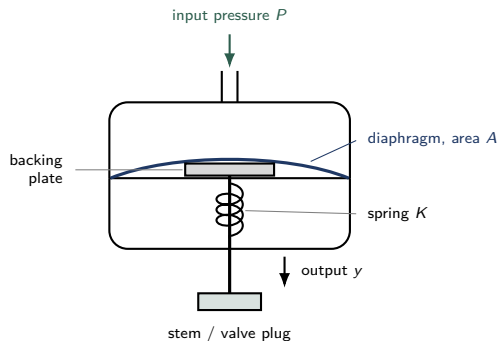
Diaphragm actuator Pressure on a diaphragm drives a stem against a return spring — the translational output.

The flapper-nozzle valve



Move the flapper nearer and the restriction rises, so the back pressure climbs towards the supply; move it away and it falls towards ambient. Cf. Nagrath Fig. 4.34, p. 117.

The diaphragm actuator — and a contrast



$$\frac{\Delta Y(s)}{\Delta P(s)} = \frac{A}{Ms^2 + fs + K}$$

Common pitfall

The hydraulic cylinder has a free $1/s$. This does **not**.

The difference is the **spring**: with one, a constant pressure gives a constant *position*; without one, a constant flow gives a constant *velocity*.

Read the physics, not the picture.

Choosing between them

	Electric	Hydraulic	Pneumatic
Medium Power/weight	electric current moderate	incompressible oil very high	compressible air low
Speed	fast	fastest under load	slow (compressibility)
Stiffness	moderate	very high	low
Typical use	instruments, servos, drives	machine tools, steering, aircraft controls, presses	process control valves
Drawback	limited torque density	leaks, sealing, contamination, noise	long time delays, low stiffness
Safety	sparks	fire risk from oil mist	non-inflammable, intrinsically safe

Key idea

Not a list to memorise — three trade-offs to reason with. **Power density** pushes you to hydraulics, **safety and cost** to pneumatics, **precision and convenience** to electrics. Almost every real selection is one of those three arguments winning.

One shape, over and over

Collect what the survey produced

The *physics* of these devices have almost nothing in common — current and flux, oil under pressure, compressed air — and yet:

Device	Shape	Why
d.c. servomotor	$\frac{K}{s(\tau s + 1)}$	voltage commands <i>speed</i>
a.c. servomotor	$\frac{K}{s(\tau s + 1)}$	control voltage commands <i>speed</i>
Hydraulic transmission	$\frac{K}{s(\tau s + 1)}$	stroke angle commands <i>motor speed</i>
Hydraulic cylinder	$\frac{K}{s(\tau s + 1)}$	valve opening commands <i>piston velocity</i>
Potentiometer, synchro, tachogenerator, bellows	pure gain	they <i>report</i> ; nothing accumulates
Pneumatic diaphragm	$\frac{A}{Ms^2 + fs + K}$	the spring fixes a <i>position</i> per pressure

One shape

Key idea

Almost every **positional actuator** in control engineering, whatever its physics, has

$$G(s) = \frac{K}{s(\tau s + 1)}.$$

The $1/s$ is there because **the input commands a rate** and the output of interest accumulates it. The $1/(\tau s + 1)$ is there because **inertia and friction resist** that rate.

Sensors and error detectors are usually **pure gains** — they report, they do not accumulate.

The exception proves the rule: put a spring on the output and the accumulation stops, and with it the integrator.

Why that matters for the next eleven weeks

- Put $K/[s(\tau s + 1)]$ in a unity-feedback loop and you get a **second-order system** — *Week 5*, and its ζ and ω_n .
- The free integrator makes the open loop **type 1**: zero steady-state error to a step, finite error to a ramp — *Week 6*.
- Its pole-zero pattern — one pole at the origin, one on the negative real axis — gives the most common **root locus** you will ever sketch — *Week 8*.
- Its **Bode plot** is a -20 dB/decade line plus a single corner — *Week 10*.

Key idea

If you take one thing from today: **you have already met the plant you will study for the next eleven weeks.** The hardware survey was not a digression before the real course — it was the course, in physical form.

Where to go next

If you want the mathematics

week-01-notes.pdf derives everything stated here — the synchro coil voltages (§5), the linearisation of the a.c. servomotor (§6), the rate-feedback closed loop (§7), the hydraulic flow balances (§8), the pneumatic force balances (§9).

Primary hardware source: **Nagrath & Gopal, 2nd ed., Ch. 4** (pp. 84–121), plus §5.4 p. 137 for the potentiometer servomechanism.

Two boundaries

Components are survey depth in FEE3411 — describe working principles; you are not expected to design a servomotor or select a valve.

FEE3411 owns analysis; FEE3412 owns design, and FEE3412 contains a full hardware block that treats all of this properly.